

Quotient Space

Def. Let $(X, \mathcal{C})$ be a Topological Space, and Y be a set.

For an onto map $f: X \rightarrow Y$, define a Quotient Topology on Y induced by f as

$$\mathcal{C}_Q \stackrel{\text{def}}{=} \{U \subseteq Y \mid f^{-1}[U] \text{ is open of } (X, \mathcal{C})\}.$$

It is actually a Topology on Y , because: clearly, $\emptyset, Y \in \mathcal{C}_Q$ and

$$U_\alpha \in \mathcal{C}_Q \ (\alpha \in I) \Rightarrow f^{-1}\left[\bigcup_{\alpha \in I} U_\alpha\right] = \bigcup_{\alpha \in I} f^{-1}[U_\alpha] \text{ is also open in } X, \text{ so is } \bigcup U_\alpha.$$

Similarly, $f^{-1}\left[\bigcap_{\alpha \in I} U_\alpha\right] = \bigcap_{\alpha \in I} f^{-1}[U_\alpha]$ also open if I is finite, so is $\bigcap U_\alpha$.

Moreover, it is the finest Topology on Y such that f is Continuous, because

$$f: X \rightarrow Y \text{ is continuous iff } \forall U \in \mathcal{C}_Y, f^{-1}[U] \in \mathcal{C}_X \text{ iff } \mathcal{C}_Y \subseteq \mathcal{C}_Q.$$

where $\mathcal{C}_Y$ is arbitrary Topology on Y . $\square$

Def. Let X be a Topological Space, and $\sim$ be an equivalence relation on X .

Define the Canonical map on X as

$$\pi: X \rightarrow X/\sim : x \mapsto [x]$$

The $(X/\sim, \mathcal{C}_\pi)$ is called Quotient Space where $\mathcal{C}_\pi$ is a Quotient topology on $X/\sim$, induced by π . That is,

$$\mathcal{C}_\pi = \{U \subseteq X/\sim \mid \pi^{-1}[U] \text{ is open}\},$$

Note: For any $U \subseteq X/\sim$, $\pi^{-1}[U] = \{x \in X \mid \pi(x) \in U\} = \bigcup_{[x] \in U} [x]$

and for any $A \subseteq X$, $\pi^{-1}[\pi[A]] = \bigcup_{x \in A} [x]$.

$\stackrel{\text{11 def}}{\text{Sat}} f(A)$, f -saturation of A .

Saturation.

Prop. Let X be a Topological Space, and let $\sim$ be equivalence relation.

For $A \subset X$, $\pi[A] \subset X/\sim$ is open $\Leftrightarrow \text{Sat}(A) \subset X$ is open.

where $\pi: X \rightarrow X/\sim: x \mapsto [x]$ is a Canonical Quotient map,

and $\text{Sat}(A) = \pi^{-1}[\pi[A]] = \bigcup_{x \in A} [x]$.

Note: Generally, $A \subset \text{Sat}(A)$.

Proof. By definition of Quotient Topology,

$\pi[A]$ is open $\Leftrightarrow \pi^{-1}[\pi[A]]$ is open. $\Rightarrow$

Let X be a Topological Space, $\sim$ be an equivalence relation on X , π be a Canonical map.
 Lemma. Let Z be a Topological Space and a map $g: X/\sim \rightarrow Z$,

g is Continuous if and only if $g \circ \pi$ is Continuous.

Proof. Since $\pi: X \rightarrow X/\sim$ is Continuous, trivially $g \circ \pi$ is Continuous.

Conversely, assume $g \circ \pi$ is Continuous. Then, given an open $U \subseteq Z$,

$(g \circ \pi)^{-1}[U] = \pi^{-1}[g^{-1}[U]]$ is open of X , so $g^{-1}[U]$ is open of $X/\sim$. $\square$

Lemma. Let Z be a Topological Space.

If a Continuous map $f: X \rightarrow Z$ satisfies $x \sim y \Rightarrow f(x) = f(y)$,
 then a map induced by f :

$$\bar{f}: X/\sim \rightarrow Z: [x] \mapsto f(x)$$

is Continuous and it is a Unique map such that $\bar{f} \circ \pi = f$.

Proof. $\bar{f}$ is well-defined by: $[x] = [y] \Leftrightarrow x \sim y \Rightarrow f(x) = f(y)$.

Given $x \in X$, $f(x) = \bar{f}([x]) = \bar{f}(\pi(x)) = (\bar{f} \circ \pi)(x)$, so $\bar{f} \circ \pi = f$

By the Lemma above, $\bar{f}$ is Continuous. And, given a map $g: X/\sim \rightarrow Z$ s.t. $g \circ \pi = f$,

$$g([x]) = g(\pi(x)) = f(x) = \bar{f}([x])$$

So the Uniqueness is guaranteed.

Lemma. Let Z be a Topological Space.

If a Continuous onto map $f: X \rightarrow Z$ satisfies $x \sim y \Leftrightarrow f(x) = f(y)$ and either open or closed
 then a map induced by f :

$$\bar{f}: X/\sim \rightarrow Z: [x] \mapsto f(x)$$

is Homeomorphism.

Lemma. If a continuous onto open (or closed) map $f: X \rightarrow Y$ satisfies
 $x \sim y \Leftrightarrow f(x) = f(y)$

Then the induced map

$$\tilde{f}: X/\sim \rightarrow Z: [x] \mapsto f(x)$$

is Homeomorphism.

Proof. Notice that $f = \tilde{f} \circ \pi$. Suppose that f is an open map.

Then, given open set $U \subset X/\sim$, $\pi^{-1}[U]$ is open. And, $U = \pi[\pi^{-1}[U]]$,

$$\tilde{f}[U] = \tilde{f}[\pi[\pi^{-1}[U]]] = f[\pi^{-1}[U]] \subset Z \text{ is open, since } f \text{ is open map}$$

Similarly, given closed set $C \subset X/\sim$,

$$\tilde{f}[C] = \tilde{f}[\pi[\pi^{-1}[C]]] = f[\pi^{-1}[C]] \text{ is closed if } f \text{ is closed map. } \square$$

Quotient map.

Def. Given Topological space X and Y , a map $f: X \rightarrow Y$ is called a quotient map if:
 f is Continuous, Onto, and satisfies for any $U \subseteq Y$,

U is open in $Y \xLeftrightarrow[\text{Quotient}]{f \text{ Continuous}} f^{-1}[U]$ is open in X .

Prop. Composition of Quotient map is also Quotient map.

Proof. Let $f: X \rightarrow Y$ and $g: Y \rightarrow Z$ be Quotient maps. Given open set $U \subseteq Z$,

$(g \circ f)^{-1}[U]$ is open $\Leftrightarrow f^{-1}[g^{-1}[U]]$ is open $\Leftrightarrow g^{-1}[U]$ is open $\Leftrightarrow U$ is open

And, $(g \circ f)[X] = g[f[X]] = g[Y] = Z$, $(g \circ f)$ is a Quotient map. $\square$

Prop. If a Continuous Onto map $f: X \rightarrow Y$ is either open or closed map, then f is Quotient map.

Proof. Given $U \subseteq Y$, if $f^{-1}[U]$ is open then $f[f^{-1}[U]] = U$ is open, if f is open map.

And, if $f^{-1}[U]$ is open, then $f[X \setminus f^{-1}[U]]$ is closed if f is closed map.

Since f is onto, $X \setminus f^{-1}[U] = f^{-1}[Y \setminus f^{-1}[U]] = f^{-1}[Y \setminus U]$, so

$f[X \setminus f^{-1}[U]] = f[f^{-1}[Y \setminus U]] = Y \setminus U$ is closed, or U is open. $\square$

But, there is a quotient map neither open nor closed: define $X = \{(x, y) \in \mathbb{R}^2 \mid x \geq 0 \text{ or } y = 0\}$.

$f: X \rightarrow \mathbb{R}; (x, y) \mapsto x$

Then, it is clearly onto, and Continuous is given by: for any $a < b$ in $\mathbb{R}$,

$f^{-1}[(a, b)] = [(a, b) \times \mathbb{R}] \cap X$ is open.

To show f is quotient map, suppose that $f^{-1}[U]$ is open for some $U \subseteq \mathbb{R}$.

Given $x \in U$, $f((x, 0)) = x \in U$, so $(x, 0) \in f^{-1}[U]$. Since $f^{-1}[U]$ is open, for some $\epsilon > 0$, $(x, 0) \in B_\epsilon((x, 0)) \cap X \subseteq f^{-1}[U]$. Thus, $f[B_\epsilon((x, 0))] = (x - \epsilon, x + \epsilon) \cap f[X] = (x - \epsilon, x + \epsilon) \subseteq U$.
So, U is open.

Moreover, $B_1((0, 2)) \cap X$ is open of X , but $f[B_1((0, 2)) \cap X] = [0, 1)$, not open.

And, $F = \{(\frac{1}{n}, n) \mid n \in \mathbb{N}\}$ is closed because $F' = \emptyset$ so $F = \overline{F}$.

But, $f[F] = \{\frac{1}{n} \mid n \in \mathbb{N}\}$ is not closed in $\mathbb{R}$, since $0 \notin f[F]$ but $0 \in f[F]'$.

Comment: A projection $\pi: \mathbb{R}^2 \rightarrow \mathbb{R}$ cannot be Closed map: $F = \{(\frac{1}{n}, n) \mid n \in \mathbb{N}\}$.

Quotient map.

Def. Let X, Y be Topological spaces

A onto map $f: X \rightarrow Y$ satisfying

$$U \subseteq Y \text{ is open} \Leftrightarrow f^{-1}[U] \subseteq X \text{ is open.}$$

Note: A space X and a onto map $f: X \rightarrow Y$, let Y be equipped:

$$\mathcal{L}_Q := \{U \subseteq Y \mid f^{-1}[U] \subseteq X \text{ is open}\}.$$

Then, f is a quotient map.

Lemma. Let $f: X \rightarrow Y$ be a Continuous onto map.

Then, f is a quotient map if and only if

Given a space Z and a map $g: Y \rightarrow Z$, g is conti. $\Leftrightarrow g \circ f$ is conti.

Prop. Let X be a Topological space and $\sim$ be an equivalent relation on X .
If the Canonical Projection $\pi: X \rightarrow X/\sim: x \mapsto [x]$ is open map,
then for a basis $\mathcal{B}$ for X , $\mathcal{B}_\sim = \{\pi[B] \mid B \in \mathcal{B}\}$ is a basis for $X/\sim$.

Proof. Let $\mathcal{U} \subseteq X/\sim$ be an open set, and let $[p] \in \mathcal{U}$ be given.

Since $\mathcal{U}$ is open, $\pi^{-1}[\mathcal{U}] \subseteq X$ is open in X .

Meanwhile, for $p \in [p]$ $\pi(p) = [p] \in \mathcal{U}$, so $p \in \pi^{-1}[\mathcal{U}]$. Thus, there is $B \in \mathcal{B}$ such that

$$p \in B \subseteq \pi^{-1}[\mathcal{U}].$$

Now, since π is onto,

$$[p] = \pi(p) \in \pi[B] \subseteq \pi[\pi^{-1}[\mathcal{U}]] = \mathcal{U}$$

Since π is open map, $\pi[B]$ is open, so $\mathcal{B}_\sim$ is a basis for $X/\sim$. ~~QED~~

Note: so, if π is open, then X 2nd-countable $\Rightarrow X/\sim$ ".

Examples.

Define equivalence relation $\sim$ on $\mathbb{R}$ generated by $x \sim y \Leftrightarrow x - y \in \mathbb{Z}$.
Then, $\mathbb{R}/\sim = \mathbb{R}/\mathbb{Z} = \{a + \mathbb{Z} \mid a \in \mathbb{R}\} = \{a + \mathbb{Z} \mid a \in [0, 1)\}$.

$U \subseteq \mathbb{R}/\sim$ is open set if and only if $\pi^{-1}[U]$ is open,

where $\pi: \mathbb{R} \rightarrow \mathbb{R}/\sim; x \mapsto x + \mathbb{Z}$.

So, the topology $\mathcal{T}_\sim$ on $\mathbb{R}/\sim$ is generated by $\mathcal{B}_\sim = \{(a, b) + \mathbb{Z} \mid a < b\}$, because:

Given $(a, b) \subseteq \mathbb{R}$, $\pi[(a, b)] = (a, b) + \mathbb{Z} = \{r + \mathbb{Z} \mid a < r < b\}$,

and $\text{Sat}_\pi(a, b) = \pi^{-1}[\pi[(a, b)]] = \bigcup_{r \in (a, b)} r + \mathbb{Z} = \bigcup_{n \in \mathbb{Z}} (a + n, b + n)$.

Since the saturation is open, so $\pi[(a, b)]$ is also open. Therefore, the Canonical projection π is open map, so the $\mathcal{B}_\sim$ is basis for $\mathbb{R}/\sim$.

Moreover, $\mathbb{R}/\sim$ is Homeomorphic to S^1 , because i define a map

$$f: \mathbb{R} \rightarrow S^1; x \mapsto e^{2\pi i x}.$$

Then, f is continuous onto and open map satisfying

$$x \sim y \Leftrightarrow x - y \in \mathbb{Z} \Leftrightarrow e^{2\pi i(x-y)} = 1 \Leftrightarrow f(x) = f(y)$$

so, the induced map

$$\tilde{f}: \mathbb{R}/\sim \rightarrow S^1; [x] \mapsto f(x)$$

is well-defined and injective, and since f and π are onto and $\tilde{f} \circ \pi = f$,

$$\tilde{f}[\pi^{-1}[U]] = \tilde{f}[\pi[x]] = (\tilde{f} \circ \pi)[x] = f[x] = U.$$

so $\tilde{f}$ is also surjection. And, given open $U \subseteq S^1$, $\tilde{f}^{-1}[U]$ is open and

$$\tilde{f}^{-1}[U] \text{ is open} \Leftrightarrow \pi^{-1}[\tilde{f}^{-1}[U]] \text{ is open} \Leftrightarrow f^{-1}[U] \text{ is open}$$

Therefore $\tilde{f}$ is also continuous. Finally, since the f is open map, given open $V \subseteq \mathbb{R}/\sim$,

$$\tilde{f}[V] = \tilde{f}[\pi[\pi^{-1}[V]]] = f[\pi^{-1}[V]] \text{ is open, so}$$

Continuous open bijective map $\tilde{f}$ is Homeomorphism. $\square$

